

DD_PROV-INT-01 - Provisioning Broadcast and Reconciliation

Version: 1.0 Status: Draft for review Bundle: 06_fulfillment Audience: fulfillment developers, provisioning integration developers, NOC, QA, DevOps Related DDs:

- DD_FUL-03 - Service Activation
- DD_FUL-04 - Restriction Fulfillment
- DD_FUL-05 - Termination Fulfillment
- DD_FUL-07 - HomePass Lifecycle Fulfillment
- DD_PLM-CFG-01 - Service Catalog
- DD_RLM-CFG-01 - HomePass Configuration
- DD_WO-01 - Work Order Module
- DD_TCK-01 - Ticketing and Case Management
- FE-APP-01 - BSS Backoffice Web App

1. Purpose

PROV-INT-01 defines the shared BSS provisioning integration layer used to send commands to network and service platforms, track command execution, retry safely, reconcile actual network state with BSS desired state, and perform controlled force-sync operations.

This DD closes a gap in the fulfillment design. Existing fulfillment DDs define when a service should be activated, restricted, resumed, suspended, modified, or revoked. This DD defines how the low-level command dispatch and reconciliation are tracked across all those flows.

Examples of provisioning targets:

- OLT / fiber access controller or NMS.
- HFC CMTS / DOCSIS platform.
- AAA / RADIUS / subscriber policy platform.
- VoIP switch.
- IPTV middleware.
- DHCP/IP address management system.
- Partner service activation API.

2. What This DD Owns

PROV-INT-01 owns:

- Provisioning target registry.
- Adapter configuration for each external platform.
- Common provisioning command envelope.
- Idempotent command ledger.
- Command attempts, retries, timeout handling, and final status.
- Broadcast to multiple target systems where one service action needs more than one vendor command.
- Desired network state snapshot from BSS.
- Observed network state snapshot from external platforms.
- Reconciliation runs and discrepancy queues.

- Manual force-sync requests with approval/audit.
- NOC-facing command and reconciliation APIs.

3. What This DD Does Not Own

Concern	Owner
Deciding that a subscription should activate, restrict, resume, terminate, or migrate	Subscription workflow DDs
Business fulfillment orchestration and per-service result aggregation	FUL-03, FUL-04, FUL-05, FUL-07
Service definitions, profile parameters, and provisioner_key	PLM-CFG-01
HomePass endpoint resolution, node code, and port	RLM-CFG-01
Work order execution in the field	WO-01
Equipment ownership and inventory	OSR docs
Vendor-specific command syntax	Adapter implementation code behind PROV-INT

Rule: PROV-INT-01 does not decide which package, service, or subscription state is correct. It sends and reconciles technical commands based on desired state supplied by the owning modules.

4. Design Decision

FUL-03, FUL-04, and FUL-05 remain the business fulfillment orchestrators. PROV-INT-01 is the common provisioning command and reconciliation layer behind them.

Recommended call pattern:

```
Subscription Workflow
-> FUL-03 / FUL-04 / FUL-05
-> PROV-INT command API
-> vendor adapter
-> network/service platform
```

The existing "Provisioner adapter" concept remains valid, but adapter implementations should write through the PROV-INT command ledger instead of hiding command state inside the fulfillment module. This gives NOC and support one place to inspect provisioning commands across activation, restriction, resume, termination, migration, and force-sync.

5. High Level Architecture

```
flowchart LR
    SWF["SUB-WF<br/>Activation, restriction, termination"]
    FUL["FUL-03/04/05<br/>Business fulfillment"]
    PI["PROV-INT-01<br/>Command ledger + dispatcher"]
    AD["Provider adapters<br/>GPON, HFC, VOIP, IPTV"]
    NET["Network/service platforms"]
    REC["Reconciliation worker"]
    NOC["Backoffice/NOC"]
    EVT["Kafka events"]

    SWF --> FUL
    FUL --> PI
    PI --> AD
    AD --> NET
    NET --> REC
    REC --> PI
    NOC --> PI
    PI --> EVT
```

6. Core Concepts

6.1 Provisioning Target

A provisioning target is an external system to which SOPHIX sends network or service commands.

Examples:

Target code	Meaning
HUAWEI_NCE_GPON_KE	Huawei access controller for GPON provisioning.
VOIPSWITCH_NRB_01	VoIP switch serving Nairobi subscribers.
IPTV_MIDDLEWARE_KE	IPTV middleware or entitlement platform.
DOCSIS_CMTS_KE	HFC/DOCSIS controller.

6.2 Provisioning Command

A provisioning command is one technical instruction sent to one target.

Examples:

- Create ONT service profile.
- Set internet speed profile.
- Create SIP profile.
- Bar outgoing voice.
- Suspend subscriber access.
- Delete IPTV entitlement.

6.3 Broadcast

Broadcast means one fulfillment action creates multiple provisioning commands.

Example: activating triple-play may create:

1. GPON internet profile command.
2. VoIP SIP profile command.
3. IPTV entitlement command.

6.4 Desired State

Desired state is what BSS believes should exist on the network based on subscription/package/service state.

Example:

```
{
  "subscriptionId": "SUB-700045",
  "serviceRef": "svc_INET_100M_GPON",
  "desiredStatus": "ACTIVE",
  "speedDownMbps": 100,
  "speedUpMbps": 20,
  "targetNodeCode": "OLT-NRB-KAREN-02",
  "targetPort": "0/4/07"
}
```

6.5 Observed State

Observed state is what the network platform reports during reconciliation.

Example:

```
{
  "targetCode": "HUAWEI_NCE_GPON_KE",
  "subscriberKey": "SUB-700045:INET",
  "observedStatus": "ACTIVE",
  "speedDownMbps": 50,
  "targetPort": "0/4/07"
}
```

If desired state says 100 Mbps but observed state says 50 Mbps, PROV-INT creates a reconciliation mismatch.

7. Main Runtime Flows

7.1 Service Activation Command Flow

```
sequenceDiagram
    participant F as FUL-03
    participant P as PROV-INT
    participant A as GPON Adapter
    participant N as Network Platform
    participant E as Kafka

    F->>P: POST /api/provisioning/commands
    P->>P: Validate idempotency and store command
    P->>A: Dispatch command
    A->>N: Vendor API call
    N-->>A: Vendor command accepted/succeeded
    A-->>P: Adapter result
    P-->>F: Command result
    P->>E: ProvisioningCommandSucceeded
```

7.2 Async Command Flow

Some vendor systems return only "accepted" and complete later.

```
sequenceDiagram
    participant F as FUL
    participant P as PROV-INT
    participant A as Adapter
    participant N as Vendor
    participant W as Status Worker

    F->>P: Command with executionMode=ASYNC_ACCEPTED
    P->>A: Submit vendor command
    A->>N: Submit
    N-->>A: vendorCommandId
    A-->>P: ACCEPTED
    P-->>F: ACCEPTED
    W->>N: Poll vendorCommandId
    N-->>W: COMPLETE
    W->>P: Mark SUCCEEDED
```

7.3 Reconciliation Flow

```
flowchart TD
    A["Start reconciliation run"] --> B["Load BSS desired state"]
    B --> C["Fetch observed state from target"]
    C --> D["Normalize observed state"]
    D --> E["Compare desired vs observed"]
    E --> F{"Mismatch?"}
    F --> G["Create reconciliation item"]
    F --> H["Mark matched"]
    G --> I["NOC reviews / force-sync"]
    H --> I
```

8. Business Rules

Rule ID	Rule	Notes
R-PROV-01	Every provisioning command must be stored before dispatch.	Enables retry and audit.
R-PROV-02	Command idempotency key must be unique per technical command.	Prevents duplicate network changes.
R-PROV-03	PROV-INT must not change subscription, invoice, wallet, or work-order state.	It only reports command results.
R-PROV-04	A fulfillment action may create multiple provisioning commands.	Broadcast is expected.
R-PROV-05	Retryable failures are retried with exponential backoff.	Adapter decides retryable vs final where possible.
R-PROV-06	Non-idempotent vendor commands require adapter-level read-before-write or vendor idempotency token.	Avoid duplicate profiles.

Rule ID	Rule	Notes
R-PROV-07	Force-sync requires RBAC permission and usually approval for destructive actions.	Example: deleting a live profile.
R-PROV-08	Reconciliation mismatches do not auto-fix by default.	They create review items.
R-PROV-09	Auto-fix can be enabled only per target/command type and must be audited.	Start disabled for medium deployment.
R-PROV-10	Desired state is sourced from BSS owner modules.	PROV-INT does not invent desired state.
R-PROV-11	Observed state snapshots must store the source target and collection timestamp.	Needed for NOC trust.
R-PROV-12	Adapter credentials are stored in secret manager, not database rows.	Security baseline.

9. Command Status Model

Status	Meaning
RECEIVED	Command stored but not yet dispatched.
DISPATCHING	Dispatcher is sending the command.
ACCEPTED	Vendor accepted command but final result is not known.
SUCCEEDED	Command completed successfully.
FAILED_RETRYABLE	Command failed but can be retried.
FAILED_FINAL	Command failed and should not retry without human action.
TIMED_OUT	Vendor did not respond within configured timeout.
CANCELLED	Command cancelled before dispatch.
SUPERSEDED	A newer command replaced this one before dispatch.

10. Data Model

10.1 provisioning_target

Stores external provisioning systems available to an operator.

Use this table when onboarding a network/service platform.

Column	Type	Purpose	Sample
target_id	UUID PK	Internal target id.	tgt-001
operator_code	TEXT	Operator/tenant.	WIK
target_code	TEXT	Stable target code.	HUAWEI_NCE_GPON_KE
target_type	TEXT	GPON_NMS, HFC_CMTS, AAA, VOIP_SWITCH, IPTV_MIDDLEWARE, PARTNER_API.	GPON_NMS
region_code	TEXT NULL	Region or technical area.	NRB
status	TEXT	ACTIVE, SUSPENDED, RETIRED.	ACTIVE
base_url_ref	TEXT NULL	Non-secret endpoint reference or config key.	cfg/huawei-nce-ke/base-url
secret_ref	TEXT	Secret manager reference.	secret/prov/huawei-nce-ke
capabilities_json	JSONB	Supported command types.	{"commands":["ACTIVATE_INET","SUSPEND_INET"]}
created_at	TIMESTAMP	Creation timestamp.	2026-05-31T09:00:00Z
updated_at	TIMESTAMP	Last update timestamp.	2026-05-31T09:00:00Z

Recommended indexes:

- Unique (operator_code, target_code).
- (operator_code, status, target_type).

10.2 provisioning_adapter_config

Stores adapter behavior for a target and provisioner key.

Use this table when mapping PLM provisioner_key values to actual platform adapters.

Column	Type	Purpose	Sample
adapter_config_id	UUID PK	Internal config id.	pac-001
operator_code	TEXT	Operator/tenant.	WIK
provisioner_key	TEXT	PLM service provisioner key.	gpon-res
target_id	UUID FK	Provisioning target.	tgt-001
adapter_class	TEXT	Code adapter implementation id.	HuaweiNceGponAdapter
execution_mode_default	TEXT	SYNC_REQUIRED or ASYNC_ACCEPTED.	SYNC_REQUIRED
timeout_ms	INTEGER	Per-command timeout.	25000
max_retry_count	INTEGER	Max automatic retries.	5
retry_policy_json	JSONB	Backoff policy.	{"baseSeconds":30,"factor":2}
status	TEXT	ACTIVE, SUSPENDED.	ACTIVE

Recommended indexes:

- Unique (operator_code, provisioner_key, target_id).

10.3 provisioning_command

Stores one BSS command to one provisioning target.

Use this table for operational tracking, idempotency, retry, and audit.

Column	Type	Purpose	Sample
command_id	UUID PK	Internal command id.	pcmd-001
operator_code	TEXT	Operator/tenant.	WIK
idempotency_key	TEXT	Unique command key.	FUL03:SUB-700045:svc_INET_100M:activate
source_module	TEXT	Caller module.	FUL-03
source_ref_id	TEXT	Fulfillment id or workflow id.	flb-activate-001
subscription_id	TEXT	Subscription impacted.	SUB-700045
customer_id	TEXT	Customer impacted.	CUS-100045
homepass_id	TEXT NULL	HomePass target.	HP-300045
service_ref	TEXT	PLM service ref.	svc_INET_100M_GPON
command_type	TEXT	Technical command type.	ACTIVATE_SERVICE
target_id	UUID FK	Target platform.	tgt-001
target_node_code	TEXT NULL	Network node.	OLT-NRB-KAREN-02
target_port	TEXT NULL	Port or logical endpoint.	0/4/07
payload_json	JSONB	Normalized command payload.	{"speedDownMbps":100}
execution_mode	TEXT	SYNC_REQUIRED, ASYNC_ACCEPTED.	SYNC_REQUIRED
status	TEXT	Command status.	SUCCEEDED
vendor_command_id	TEXT NULL	External command id.	NCE-CMD-8801
last_error_code	TEXT NULL	Last error.	TARGET_TIMEOUT
created_at	TIMESTAMP	Creation timestamp.	2026-05-31T09:42:00Z
updated_at	TIMESTAMP	Last update timestamp.	2026-05-31T09:42:02Z

Recommended indexes:

- Unique (operator_code, idempotency_key).
- (operator_code, status, created_at).

- (subscription_id, created_at).
- (target_id, status).

10.4 provisioning_command_attempt

Stores each dispatch attempt for a command.

Use this table when debugging timeouts, retries, and vendor errors.

Column	Type	Purpose	Sample
attempt_id	UUID PK	Internal attempt id.	pcma-001
command_id	UUID FK	Parent command.	pcmd-001
attempt_no	INTEGER	Attempt number.	1
adapter_class	TEXT	Adapter used.	HuaweiNceGponAdapter
status	TEXT	SUCCESS, FAILED_RETRYABLE, FAILED_FINAL, TIMEOUT.	SUCCESS
request_payload_redacted	JSONB	Redacted outbound payload.	{"speedDownMbps":100}
response_payload_redacted	JSONB NULL	Redacted vendor response.	{"commandId":"NCE-CMD-8801"}
vendor_status_code	TEXT NULL	Vendor status/code.	OK
duration_ms	INTEGER	Attempt duration.	1200
error_code	TEXT NULL	Stable SOPHIX error.	PROV_TARGET_TIMEOUT
created_at	TIMESTAMP	Attempt timestamp.	2026-05-31T09:42:01Z

Recommended indexes:

- (command_id, attempt_no).
- (status, created_at).

10.5 provisioning_desired_state

Stores BSS desired technical state per subscription service.

Use this table as the comparison base for reconciliation.

Column	Type	Purpose	Sample
desired_state_id	UUID PK	Internal id.	pds-001
operator_code	TEXT	Operator/tenant.	WIK
subscription_id	TEXT	Subscription.	SUB-700045
customer_id	TEXT	Customer.	CUS-100045
homepass_id	TEXT	HomePass.	HP-300045
service_ref	TEXT	PLM service.	svc_INET_100M_GPON
target_id	UUID FK	Target system expected to hold state.	tgt-001
subscriber_key	TEXT	Target lookup key.	SUB-700045:INET
desired_status	TEXT	ACTIVE, SUSPENDED, RESTRICTED, TERMINATED, NOT_PRESENT.	ACTIVE
desired_profile_json	JSONB	Expected profile fields.	{"speedDownMbps":100}
source_module	TEXT	Module that last wrote desired state.	FUL-03
source_ref_id	TEXT	Fulfillment id.	flb-activate-001
effective_from	TIMESTAMP	Desired state effective time.	2026-05-31T09:42:02Z
updated_at	TIMESTAMP	Last update timestamp.	2026-05-31T09:42:02Z

Recommended indexes:

- Unique (operator_code, subscription_id, service_ref, target_id).
- (target_id, desired_status).

10.6 provisioning_observed_state

Stores latest observed state fetched from a provisioning target.

Use this table to know what the network says right now or at last reconciliation.

Column	Type	Purpose	Sample
observed_state_id	UUID PK	Internal id.	pos-001
operator_code	TEXT	Operator/tenant.	WIK
target_id	UUID FK	Target queried.	tgt-001
subscriber_key	TEXT	Target subscriber key.	SUB-700045:INET
observed_status	TEXT	Observed platform status.	ACTIVE
observed_profile_json	JSONB	Observed profile fields.	{"speedDownMbps":50}
raw_payload_redacted	JSONB	Redacted raw target response.	{"profile":"50M"}
collected_at	TIMESTAMP	When data was fetched.	2026-05-31T23:00:00Z
source_run_id	UUID NULL	Reconciliation run id.	pr-001

Recommended indexes:

- Unique (target_id, subscriber_key).
- (target_id, collected_at).

10.7 provisioning_reconciliation_run

Stores one reconciliation run against one target or target group.

Use this table for NOC and operational reporting.

Column	Type	Purpose	Sample
run_id	UUID PK	Run id.	pr-001
operator_code	TEXT	Operator/tenant.	WIK
target_id	UUID NULL	Target reconciled, null for all targets.	tgt-001
scope_type	TEXT	FULL_TARGET, REGION, SUBSCRIPTION, SERVICE_CLASS.	FULL_TARGET
scope_value	TEXT NULL	Scope value.	NRB
status	TEXT	RUNNING, COMPLETED, FAILED, PARTIAL.	COMPLETED
desired_count	INTEGER	Desired rows compared.	10000
observed_count	INTEGER	Observed rows fetched.	9980
mismatch_count	INTEGER	Open mismatch count.	25
started_at	TIMESTAMP	Run start.	2026-05-31T23:00:00Z
completed_at	TIMESTAMP NULL	Run completion.	2026-05-31T23:15:00Z

Recommended indexes:

- (operator_code, status, started_at).
- (target_id, started_at).

10.8 provisioning_reconciliation_item

Stores one desired-vs-observed mismatch.

Use this table for NOC review, force-sync, or ticket creation.

Column	Type	Purpose	Sample
item_id	UUID PK	Item id.	pr-001
run_id	UUID FK	Reconciliation run.	pr-001
target_id	UUID FK	Target system.	tgt-001
subscription_id	TEXT NULL	Subscription if known.	SUB-700045
service_ref	TEXT NULL	Service if known.	svc_INET_100M_GPON
subscriber_key	TEXT	Target subscriber key.	SUB-700045:INET

Column	Type	Purpose	Sample
mismatch_type	TEXT	MISSING_ON_NETWORK, EXTRA_ON_NETWORK, STATUS_MISMATCH, PROFILE_MISMATCH, PORT_MISMATCH.	PROFILE_MISMATCH
severity	TEXT	LOW, MEDIUM, HIGH, CRITICAL.	HIGH
desired_snapshot_json	JSONB	Desired state.	{"speedDownMbps":100}
observed_snapshot_json	JSONB	Observed state.	{"speedDownMbps":50}
status	TEXT	OPEN, IN_REVIEW, RESOLVED, IGNORED, TICKET_CREATED.	OPEN
resolution_notes	TEXT NULL	NOC notes.	Force-sync approved
created_at	TIMESTAMP	Creation timestamp.	2026-05-31T23:15:00Z
resolved_at	TIMESTAMP NULL	Resolution timestamp.	NULL

Recommended indexes:

- (status, severity, created_at).
- (subscription_id, status).
- (target_id, mismatch_type, status).

10.9 provisioning_force_sync_request

Stores manual or approved corrective action requests.

Use this table when NOC decides to push BSS desired state to the network or align BSS to confirmed network state.

Column	Type	Purpose	Sample
force_sync_id	UUID PK	Force-sync request id.	pfs-001
operator_code	TEXT	Operator/tenant.	WIK
source_item_id	UUID NULL	Reconciliation item that triggered it.	pri-001
subscription_id	TEXT	Subscription.	SUB-700045
service_ref	TEXT	Service.	svc_INET_100M_GPON
target_id	UUID FK	Target system.	tgt-001
sync_direction	TEXT	BSS_TO_NETWORK, NETWORK_TO_BSS, MARK_IGNORE.	BSS_TO_NETWORK
requested_action	TEXT	Corrective command.	REAPPLY_PROFILE
status	TEXT	PENDING_APPROVAL, APPROVED, RUNNING, COMPLETED, FAILED, CANCELLED.	PENDING_APPROVAL
requested_by_user_id	TEXT	Requesting user.	usr-noc-001
approved_by_user_id	TEXT NULL	Approver user.	usr-noc-lead
command_id	UUID NULL	Corrective provisioning command.	pcmd-999
reason	TEXT	Why action is needed.	Customer on wrong speed profile
created_at	TIMESTAMP	Creation timestamp.	2026-05-31T23:20:00Z
updated_at	TIMESTAMP	Last update timestamp.	2026-05-31T23:21:00Z

Recommended indexes:

- (status, created_at).
- (subscription_id, status).

11. API Contract

11.1 Create Provisioning Command

Called by FUL modules or approved force-sync logic.

```
POST /api/provisioning/commands
Idempotency-Key: FUL03:SUB-700045:svc_INET_100M:activate
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "sourceModule": "FUL-03",
  "sourceRefId": "flb-activate-001",
  "subscriptionId": "SUB-700045",
  "customerId": "CUS-100045",
  "homepassId": "HP-300045",
  "serviceRef": "svc_INET_100M_GPON",
  "provisionerKey": "gpon-res",
  "commandType": "ACTIVATE_SERVICE",
  "targetNodeCode": "OLT-NRB-KAREN-02",
  "targetPort": "0/4/07",
  "executionMode": "SYNC_REQUIRED",
  "payload": {
    "equipmentId": "EQP-ONT-42",
    "speedDownMbps": 100,
    "speedUpMbps": 20,
    "subscriberKey": "SUB-700045:INET"
  }
}
```

Success response:

```
{
  "commandId": "pcmd-001",
  "status": "SUCCEEDED",
  "vendorCommandId": "NCE-CMD-8801",
  "adapterRefs": {
    "provisioningProfileId": "prf-100m-v3"
  }
}
```

11.2 Read Command

```
GET /api/provisioning/commands/{command_id}
```

11.3 Retry Command

```
POST /api/provisioning/commands/{command_id}/retry
```

Only allowed for FAILED_RETRYABLE, TIMED_OUT, or approved NOC override.

11.4 Reconciliation APIs

```
POST /api/provisioning/reconciliation-runs
GET /api/provisioning/reconciliation-runs
GET /api/provisioning/reconciliation-runs/{run_id}
GET /api/provisioning/reconciliation-items?status=OPEN
POST /api/provisioning/reconciliation-items/{item_id}/resolve
```

11.5 Force-Sync APIs

```
POST /api/provisioning/force-sync-requests
POST /api/provisioning/force-sync-requests/{force_sync_id}/approve
POST /api/provisioning/force-sync-requests/{force_sync_id}/execute
POST /api/provisioning/force-sync-requests/{force_sync_id}/cancel
```

Force-sync APIs require RBAC permission and audit. Destructive actions should use EM-CFG-04 approval.

12. Events

Event	Topic	When emitted
ProvisioningCommandReceived	sophix.provisioning.command.received	Command stored.
ProvisioningCommandSucceeded	sophix.provisioning.command.succeeded	Command succeeded.
ProvisioningCommandFailed	sophix.provisioning.command.failed	Command failed final or retryable.
ProvisioningCommandTimedOut	sophix.provisioning.command.timedout	Timeout reached.
ProvisioningReconciliationCompleted	sophix.provisioning.reconciliation.com pleted	Reconciliation run completed.
ProvisioningMismatchDetected	sophix.provisioning.mismatch.detected	Mismatch item created.

Event	Topic	When emitted
ProvisioningForceSyncRequested	sophix.provisioning.forcesync.requested	Force-sync request created.
ProvisioningForceSyncCompleted	sophix.provisioning.forcesync.completed	Force-sync completed.

Sample ProvisioningMismatchDetected:

```
{
  "eventType": "ProvisioningMismatchDetected",
  "operatorCode": "WIK",
  "itemId": "pri-001",
  "subscriptionId": "SUB-700045",
  "serviceRef": "svc_INET_100M_GPON",
  "targetCode": "HUAWEI_NCE_GPON_KE",
  "mismatchType": "PROFILE_MISMATCH",
  "severity": "HIGH"
}
```

13. Frontend Usage

13.1 FE-APP-01 Backoffice

Backoffice uses PROV-INT-01 for:

- Provisioning command search.
- Command detail with attempts and vendor references.
- Retry command action.
- Reconciliation run dashboard.
- Open mismatch queue.
- Force-sync request creation and approval.
- Deep links from customer 360, subscription detail, ticket detail, and NOC screens.

13.2 FE-APP-03 Contractor Mobile App

The contractor app normally does not call PROV-INT directly.

It completes work orders in WO-01. The subscription workflow or fulfillment module then triggers provisioning. The mobile app may show provisioning result read-only if the WO completion response includes it.

13.3 FE-APP-04 Customer Self Care

Self-care does not call PROV-INT directly.

If activation is delayed due to provisioning, self-care reads customer-friendly status from subscription/order/ticket APIs, not raw provisioning command errors.

14. Infra Recommendation

14.1 Medium Budget Deployment

Recommended starting point:

Component	Recommendation
API pods	2 pods, 1 vCPU, 1.5-2 GB RAM each.
Dispatcher worker	1-2 pods, 1 vCPU, 2 GB RAM each.
Reconciliation worker	1 pod, 2 vCPU, 4 GB RAM for nightly/batch runs.
DB	Shared fulfillment PostgreSQL schema prov_int; partition command/attemp tables monthly.
Kafka	Shared Kafka for events and optional command queue.

Component	Recommendation
Redis	Optional short-lived locks only. DB remains source of truth.

14.2 Recommended Partitions

Partition by month:

- provisioning_command
- provisioning_command_attempt
- provisioning_observed_state if storing historical snapshots
- provisioning_reconciliation_item

Keep provisioning_desired_state as current-state table with history in audit or event log.

14.3 OS Recommendation

Use Ubuntu Server 24.04 LTS for VPS deployments or hardened Debian/Ubuntu slim base images for containers.

Provisioning workers should run on private subnets only. They need outbound access to network management systems, but they should not be internet-facing.

15. AWS Deployment Recommendation

AWS component	Usage
ECS/EKS	Run PROV-INT API, dispatcher, and reconciliation workers.
RDS PostgreSQL	Command ledger and reconciliation tables.
MSK or managed Kafka equivalent	Events and async dispatch jobs.
Secrets Manager	Vendor API credentials and certificates.
VPC private subnets	Workers and network connectivity to target systems.
Site-to-Site VPN or Direct Connect	Connectivity to NMS/OSS platforms if outside AWS.
CloudWatch	Logs, metrics, alarms.

Required alarms:

- Command failure rate above threshold.
- Retry backlog growing.
- No reconciliation run completed for a target within expected window.
- Critical mismatch count above threshold.
- Force-sync failures.
- Target unavailable for more than configured duration.

16. Implementation Notes

1. Start with command ledger and synchronous dispatch before building auto-repair.
2. Use idempotency keys from FUL correlation ids and service refs.
3. Keep adapter classes thin and testable.
4. Normalize vendor responses into stable SOPHIX statuses.
5. Do not expose vendor errors directly to customers.
6. Store redacted request/response payloads for support.
7. Reconciliation should initially create review items, not auto-fix.
8. Force-sync actions should create normal provisioning_command rows so audit is consistent.

17. Definition Of Done

PROV-INT-01 is complete when:

- Provisioning target and adapter config are implemented.
- FUL-03 can create commands through PROV-INT for activation.
- FUL-04 can create commands through PROV-INT for restriction/suspension.
- FUL-05 can create commands through PROV-INT for termination.
- Command idempotency prevents duplicate vendor actions.
- Retry and timeout handling are tested.
- Command detail and attempt logs are available to Backoffice.
- Reconciliation can compare at least one target's observed state to BSS desired state.
- Mismatches create actionable items.
- Force-sync is implemented with RBAC and audit.